

67. If a vector of magnitude 2 units makes an angle $\frac{\pi}{3}$ with $2\hat{i}$, $\frac{\pi}{4}$ with $3\hat{j}$ and an acute angle θ with $4\hat{k}$, then what are the components of the vector ?

- (a) $(1, \sqrt{2}, 1)$
- (b) $(1, -\sqrt{2}, 1)$
- (c) $(1, -\sqrt{2}, -1)$
- (d) $(1, \sqrt{2}, -1)$

68. Consider the following in respect of moment of a force :

1. The moment of force about a point is independent of point of application of force.
2. The moment of a force about a line is a vector quantity.

Which of the statements given above is/are correct ?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

69. For any vector $\vec{r}$, what is

$(\vec{r} \cdot \hat{i})(\vec{r} \times \hat{i}) + (\vec{r} \cdot \hat{j})(\vec{r} \times \hat{j}) + (\vec{r} \cdot \hat{k})(\vec{r} \times \hat{k})$ equal to ?

- (a) $\vec{0}$
- (b) $\vec{r}$
- (c) $2\vec{r}$
- (d) $3\vec{r}$

70. Let $\vec{a}$ and $\vec{b}$ are two vectors of magnitude 4 inclined at an angle $\frac{\pi}{3}$, then what is the angle between $\vec{a}$ and $\vec{a} - \vec{b}$?

- (a) $\frac{\pi}{2}$
- (b) $\frac{\pi}{3}$
- (c) $\frac{\pi}{4}$
- (d) $\frac{\pi}{6}$

71. Let $y_1(x)$ and $y_2(x)$ be two solutions of the differential equation $\frac{dy}{dx} = x$. If $y_1(0) = 0$ and $y_2(0) = 4$, then what is the number of points of intersection of the curves $y_1(x)$ and $y_2(x)$?

- (a) No point
- (b) One point
- (c) Two points
- (d) More than two points

72. The differential equation, representing the curve $y = e^x(a \cos x + b \sin x)$ where a and b are arbitrary constants, is

- (a) $\frac{d^2y}{dx^2} + 2y = 0$
- (b) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$
- (c) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$
- (d) $\frac{d^2y}{dx^2} + y = 0$

73. यदि $f(x) = ax - b$ और $g(x) = cx + d$ इस प्रकार हैं कि $f(g(x)) = g(f(x))$, तो निम्नलिखित में से कौन-सा सही है ?

- (a) $f(d) = g(b)$
- (b) $f(b) + g(d) = 0$
- (c) $f(a) + g(c) = 2a$
- (d) $f(d) + g(b) = 2d$

74. $\int_{-1}^1 (3\sin x - \sin 3x) \cos^2 x dx$ किसके बराबर है ?

- (a) $-\frac{1}{4}$
- (b) 0
- (c) $\frac{1}{2}$
- (d) $\frac{1}{4}$

75. अवकल समीकरण

$$\left\{ 2 - \left(\frac{dy}{dx} \right)^2 \right\}^{0.6} = \frac{d^2 y}{dx^2} \text{ की क्रमशः कोटि}$$

और घात क्या हैं ?

- (a) 2, 2
- (b) 2, 3
- (c) 5, 2
- (d) 2, 5

76. यदि $\frac{dy}{dx} = 2e^x y^3$, $y(0) = \frac{1}{2}$ तो

$4y^2(2 - e^x)$ किसके बराबर है ?

- (a) 1
- (b) 2
- (c) 3
- (d) 4

77. मान लीजिए $p = \int_a^b f(x) dx$ और

$$q = \int_a^b |f(x)| dx \text{ है। यदि } f(x) = e^{-x}, \text{ तो}$$

निम्नलिखित में से कौन-सा सही है ?

- (a) $p = 2q$
- (b) $p = -q$
- (c) $4p = q$
- (d) $p = q$

78. $\int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x} dx$ किसके बराबर है ?

- (a) $\frac{\pi}{4}$
- (b) $\frac{\pi}{2}$
- (c) 1
- (d) 0

79. b का वह ऋणेतर मान क्या है जिसके लिए

$$\text{फलन } \frac{16x^3}{3} - 4bx^2 + x \text{ का } x > 0 \text{ परिसर में}$$

न तो अधिकतम मान है और न ही न्यूनतम मान है ?

- (a) $0 < b < 1$
- (b) $1 < b < 2$
- (c) $b > 2$
- (d) $0 \leq b < 1$